# AUTUMN END SEMESTER EXAMINATION-2018

## 5th Semester B.Tech & B.Tech Dual Degree

## ENGINEERING ECONOMICS

## HS-2002

**[For 2017(L.E.), 2016 & Previous Admitted Batches]**

Time: 3 Hours Full Marks: 60

***Answer any Six questions including question No.1 which is compulsory.***
*The figures in the margin indicate full marks.*
*Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.*

1. (a) Diagrammatically show the shut down point of a perfectly competitive firm? [2×10]

(b) The demand curve for bread is given by Q = 500-10P.If the price increases from ₹10 to ₹15, what is the arc elasticity of demand?

(c) Explain why two Iso-Quants won't intersect with each other.

(d) Two goods X and Y are available for ₹ 5/kg and ₹ 4/kg respectively. If the consumer has an income of ₹ 200, write the budget equation and draw the budget line.

(e) Explain the relationship among average revenue, marginal revenue and price elasticity of demand.

(f) If the demand curve for product B shifts to the right as the price of product A declines, what can be inferred pertaining to the relationship between A and B?

(g) Suppose the price elasticity coefficients of demand are 2.3, 0.40, 1.27, 0.77 and 2.78 for demand schedules D1, D2, D3, D4 and D5 respectively. A one percent increase in price will result in an increase in total revenue for which of the above mentioned demand schedule/s?

(h) A driver wishes to buy gasoline and have his car washed. The car wash costs ₹1 when he buys up to 19 gallons of gas at ₹1.10 each. However if he buys 20 gallons, the car wash is free. What is the marginal cost of the twentieth gallon to him?

(i) A Company has purchased an equipment whose first cost is ₹100000 with an estimated life of eight years. The estimated salvage value of the equipment at the end of its lifetime is ₹ 20000. Determine the depreciation charge and book value for $4^{th}$ year using the Straight Line Method of Depreciation.

(j) A Hotel can have maximum 1200 tourists accommodated in it at a time. The daily fare is ₹ 700/-. The Variable Cost per tourist is ₹125/- while the fixed cost is ₹150, 000/-. Find the Break-Even number of tourists.

2. (a) Wheeler Corporation is a publisher of paperback fiction. [4]

The company's economic analyst derives a demand curve for the product of the company as

$$Q = 10{,}000 - 500P_X + 5M + 5P_Y$$

Where Q is the demand for the firm's novels

$P_X$ is the price charged for the firm's novels

M is the per capita income

$P_Y$ is the price of paperbacks by a competing publisher

The initial values of $P_X$ = ₹50, M= 5000 and $P_Y$ = ₹60.

Using this information,

Determine what effect a price increase would have on total revenues.

(b) Using the information given in Q2 (a), What will happen when the competing publisher reduces its price to ₹55? [4]

3. (a) Suppose the firm has the following total cost (TC) function: [4]

$$TC = 45200 + 1.5Q + 2.7Q^2 - 0.004Q^3$$

Write equations for the average fixed cost and the average variable cost.

What will be the value of the short-run average cost when $Q = 80$ and $Q = 70$?

(b) With the TC function as given in Q3 (a), find the Marginal cost for $Q = 50$. Establish the relationship between Average cost and marginal cost curves. [4]

4. (a) A Producer is experiencing the Production function $Q=50L^{0.5}K^{0.5}$. Given that the Price of K is 4 and that of L is 5. The Producer is currently producing 300 output using the factor combination K=20 and L=6. Is this an efficient Combination? [4]

(b) Explain the following concepts related to national income - GDP, NDP, GNP and NNP both at Factor Cost and Market Price. Which of these eight concepts truly reflects the National Income of an Economy? [4]

5. (a) A perfectly competitive firm faces a price of ₹14 per unit. It hasthe following short run cost schedule: [4]

| Output | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Total Cost | 10 | 18 | 24 | 30 | 38 | 50 | 64 | 91 | 120 |

Find the marginal cost for each level of output and also find the profit maximizing output.

(b) If the total cost (TC) function for a monopolist is given by the equation: $TC = 500 + 25Q^2$ and the demand function for the good produced by the monopolist is given by $Q = 25 - 0.05P$, find [4]

(i) The profit maximizing price and quantity, and

(ii) Maximized profit of the monopolist.

6. (a) A Company has identified two mutually exclusive investment proposals. The details of both the Proposals are given in the following table. Find the better alternative on the basis of rate of return method of comparison. [4]

| Particulars | Alternative | |
|---|---|---|
| | A | B |
| Investment (₹) | 210000 | 255000 |
| Annual Net Income (₹) | 58260 | 69000 |
| Life | 5 yrs | 5 yrs |
| Salvage Value | Nil | Nil |
| Minimum Attractive rate of return | 10% | |

(b) A firm invests in one of the two mutually alternatives the details of which are given below. Determine the better alternative based on the annual equivalent method by assuming i = 20%, compounded annually. [4]

| Particulars | Alternative A | Alternative B |
|---|---|---|
| Life | 15 Yrs | 15 Yrs |
| i | 20% | 20% |
| First Cost | ₹ 100000 | ₹110000 |
| Annual Equal Returns | ₹ 70000 | ₹ 80000 |
| Salvage Value | ₹ 10000 | ₹ 20000 |

7. (a) Details of two alternatives are given below. Select one between the two on the basis of future worth method. [4]

| Particulars | Alternative-1 | Alternative-2 |
|---|---|---|
| Initial Cost (₹) | 400000 | 800000 |
| Useful Life (yr.) | 4 | 4 |
| Salvage Value (₹) | 200000 | 550000 |
| Annual Cost (₹) | 40000 | Nil |
| i (Nominal Interest Rate) | 12% | 12% |

(b) A State govt. is planning a hydroelectric Project for a river basin. In addition to the production of electric power, this project will provide flood control, irrigation and recreation benefits. The estimated benefits and costs that are expected to be derived from this project are as follows. [4]

| Particulars | Amount (in ₹) |
|---|---|
| Life of the Project | 40 Years |
| Annual operating and maintenance costs | 3500000 |
| Initial Cost | 85000000 |
| Annual irrigation benefits | 5500000 |
| Annual Power Sales | 6500000 |
| Annual recreation benefits | 2500000 |
| Annual flood control savings | 3500000 |

Check whether the State govt. should implement the Project (assume i = 12%)

8. (a) A Company wants to purchase a machine. There are two machines A and B available in the market details of which are given below. Decide which machine should the company purchase on the basis of pay-back period method. [4]

| Year | Cash Flow for Machine A (₹) | Cash Flow for Machine B (₹) |
|---|---|---|
| 0 | -600000 | -600000 |
| 1 | 150000 | 95000 |
| 2 | 250000 | 200000 |
| 3 | 400000 | 300000 |
| 4 | 425000 | 450000 |
| 5 | 450000 | 475000 |
| 6 | 500000 | 500000 |

(b) A Company has purchased an equipment whose first cost is ₹100000 with an estimated life of eight years. The estimated salvage value of the equipment at the end of its lifetime is ₹20000. Calculate the depreciation and the book value for period 6 using the declining balance method of depreciation by assuming 0.2 for K. [2]

(c) Explain how the fiscal policy helps in controlling inflation. [2]

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